

PocketSmart – CORS and Static Routing

Member Assignment – Basila

1. Topic

CORS (Cross-Origin Resource Sharing) and Static Routing

Epic: Backend Networking and API Configuration

2. Project Overview

PocketSmart is an AI-powered app that helps users track spending, set budgets, and get personalized money-saving suggestions. The backend needs secure communication between the frontend and API, while static routing provides a defined path for network traffic between required services. CORS and static routing help the PocketSmart system communicate reliably across its application components.

3. Objective

To configure CORS so that approved frontend applications can access the PocketSmart API, and to configure static routing so network traffic can reach the required backend services through defined routes.

4. Work Steps

- 1 Identify the frontend origin(s) that need access to the PocketSmart backend.
- 2 Configure CORS in the backend to allow only the required origins.
- 3 Allow the required HTTP methods such as GET, POST, PUT, and DELETE.
- 4 Configure required request headers and credentials settings when needed.
- 5 Test API requests from the frontend and verify that valid origins are accepted.
- 6 Check that unauthorized or unapproved origins are blocked by the CORS policy.
- 7 Identify the network destinations that require fixed routes.
- 8 Create static routes that define the next hop or interface for required traffic.
- 9 Verify routing connectivity between the application components.
- 10 Test the configuration and add suitable error handling or logging for connection failures.

5. Main Parts of the Setup

- Frontend and backend origin identification
- CORS configuration and allowed origins
- Allowed HTTP methods and request headers
- Credentials and secure cross-origin access
- CORS testing and error handling
- Network destination identification
- Static route configuration
- Next-hop or network interface selection
- Connectivity and routing verification

6. Expected Outcome

The PocketSmart frontend should be able to communicate with the backend only through approved cross-origin rules, while static routes should direct required network traffic to the correct destinations. The configuration should support reliable API communication and easier troubleshooting.

7. Key Benefits

- Controls which frontend origins can access the API.
- Improves API security by limiting unnecessary cross-origin requests.
- Provides predictable paths for selected network traffic.
- Helps maintain reliable communication between PocketSmart services.
- Makes network configuration and troubleshooting easier.